

LAB 1 — Software Failure Forensics & Engineering Ethics

T6 – Toyota Unintended Acceleration

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Casual Chain

1. [Business Decision]

Toyota started cost-cutting and rapid expansion and prioritized production over safety.

2. [Process Failure]

Customer complaints of sudden acceleration were dismissed as "driver error" or "floor mat interference." Independent, rigorous software audits were bypassed.

3. [Engineering Defect]

ETCS-i software was "spaghetti code" with over 10,000 global variables, single points of failure (task death), no software/hardware Brake Override System (Smart Pedal) to prioritize brakes over the throttle.

4. [Loss]

89 fatalities occurred, over 9 million vehicles were recalled.

Failure Timeline

1. **2002–2004:** Toyota introduces ETCS-i across major vehicle lines. NHTSA complaints regarding Sudden Unintended Acceleration (SUA) begin to rise.
2. **2007:** Toyota issues a limited recall, attributing SUA entirely to heavy all-weather floor mats trapping the pedal.
3. **August 2009:** An off-duty highway patrolman and his family are killed in a Lexus ES 350 in San Diego after the throttle sticks wide open, captured on a harrowing 911 call.
4. **Late 2009 / Early 2010:** Toyota issues massive recalls for floor mats and a mechanical "sticky pedal" defect, still denying electronic/software faults.
5. **2011:** NASA/NHTSA report concludes they cannot find an electronic cause, pointing back to mechanical issues.
6. **2013:** *Bookout v. Toyota* trial. Independent software experts (Barr/Koopman) analyze Toyota's source code, revealing catastrophic architectural flaws ("spaghetti code", RTOS task death). The jury finds Toyota liable with "reckless disregard."
7. **2014:** Toyota pays a \$1.2 billion DOJ criminal penalty for intentionally concealing safety information from the public and regulators.